

William Park

236-996-9116 | www.williampark.org | linkedin.com/in/parkwilliamy | github.com/parkwilliamy | w5park@uwaterloo.ca

EDUCATION

University of Waterloo

Bachelor of Applied Science in Honours Electrical Engineering
Cumulative GPA: 91.09%

Waterloo, ON

Sept 2024 – May 2029

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, VHDL, Tcl, Bash, Assembly, Python, C/C++, Rust

Developer Tools: Git, Vivado, Quartus Prime, IcarusVerilog, Verilator, gem5, Xschem, SPICE, Ngspice, Magic, OpenOCD, GDB, Cargo, ROS2, Docker, WSL, Linux, NVIDIA JetPack

Libraries: cocotb, CUDA, MediaPipe, OpenCV, pyserial

Protocols: SPI, I2C, CAN, UART, AXI4

WORK EXPERIENCE

University of Waterloo

Undergraduate Research Co-op

May 2026 – Present

Waterloo, ON

- Researching hardware prefetcher designs for **RISC-V Vector (RVV)** indexed memory accesses, targeting irregular access patterns in vectorized workloads
- Extending **gem5's** RVV simulation infrastructure with a custom cache hierarchy, adding configurable prefetcher attachment at L1 and L2 caches for fast architectural exploration
- Implementing the prefetcher design in SystemVerilog for cycle-accurate validation against the gem5 model

UWASIC

Digital Team Lead

Oct 2025 – Present

Waterloo, ON

- Leading a team of ten students to design and fabricate the first programmable ray-tracing ASIC on the **IHP SG13G2 130nm process** through Tiny Tapeout
- Designing architecture to efficiently implement essential ray-tracing operations like division and square roots
- Conducted parameter sweeps for samples-per-pixel and color depth in a C++ ray tracer to establish performance constraints and design requirements
- Analyzed renders to measure arithmetic operation frequency, determining critical targets for hardware optimization

Kiwi Charge

Electrical Engineering Intern

Sept 2025 – Dec 2025

Toronto, ON

- Developed a ROS2-based teleoperation pipeline for a four-wheel differential-drive prototype, integrating joystick input and real-time motor command streaming with the **CAN** bus protocol
- Designed an electrical schematic for the prototype, including power distribution, motor driver interfaces, CAN bus communication, and sensor integration, enabling reliable bring-up and debugging
- Implemented Python scripts to configure and tune multiple motor drivers in parallel over CAN, **reducing setup time by 98%** and improving system scalability

PROJECTS

R500

June 2025 - Present

- Designed and implemented a **RISC-V** processor in **SystemVerilog** on the Xilinx Artix-7 FPGA
- Reached **100% compliance** with the RV32I ISA based on the official RISC-V compliance test suite
- Achieved a throughput of **46.3 million instructions per second** using a **57 MHz** clock, along with a **CPI of 1.23** and a **branch prediction accuracy of 96.5%**
- Optimized performance through pipelining, forwarding, stalling, flushing, and gshare global branch prediction
- Created unit testbenches in SystemVerilog, employing directed tests and constrained random verification
- Utilized **Tcl** scripts to automate simulation, synthesis, and static timing analysis in Vivado
- Designing split L1 instruction and data caches with **AXI4** interfaces
- Implementing precise exception and interrupt handling with CSR support